Applied Economics

Panel Data Methods

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Plan for this Lecture

1. What is Panel Data?
2. Within and Between Variation
3. Panel Data Estimators
    - Dummy Variable Estimator
    - First Differences
    - Within-Estimator

Motivation

Endogeneity due to OVB is a pervasive issue.

One approach to address it: instrumental variable regressions

Finding suitable instruments is very hard.

Today we consider an alternative approach to address certain forms of omitted variables: **panel data** allows to control for all *time-invariant* unobservables.

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Panel data: same cross-sectional unit is observed over multiple time periods

- *cross-sectional unit* can be a person, company, city, state, etc. $(i = 1, \ldots n)$
- *time period* can be a year, a month, a day, etc. $(t = 1, \ldots, T)$

We denote a sample of panel observations as $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n$, and $t = 1, ..., T$.

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Balanced panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$ $i = 1, ..., n; t = 1, ..., T$ [n x T observations in total]
Unbalanced panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n; t = \underline{t}_i ..., \overline{t}_i$, where $\underline{t}_i$ and $\overline{t}_i$ may vary across i. $\rightarrow$ we do not observe the same number of periods for each i.

Unbalancedness is prevalent in practice, but we will abstract from this for the sake of exposition.

(Unbalanced) Panel Data: Example

state	year	fatalities	seatbelt	income	min age
AK	1990	0.025	0.450	21073	16
AK	1991	0.025	0.660	21496	16
AK	1992	0.028	0.660	22073	16
AK	1993	0.030	0.690	22711	16
AL	1988	0.026	0.290	13288	18
AL	1989	0.025	0.380	14266	18
AL	1990	0.026	0.440	15213	18
AL	1991	0.026	0.470	15895	18
AL	1992	0.023	0.580	16817	18
AL	1993	0.022	0.550	17398	18
AR	1988	0.032	0.301	12637	21
AR	1990	0.029	0.340	14025	21
AR	1991	0.028	0.520	14766	21
	...	...			

i = state

t = year

Y_{it} = number of fatalities per million traffic miles

X_{1it} = seat belt usage rate

X_{2it} = median income

X_{3it} = minimum drinking age

What is Panel Data?

Our focus is on large micro-level survey data sets. These are often called "large n, small T" panels.

- many cross-sectional units n (which may be randomly sampled)
- observed for small number of time periods T (which are typically ordered)

We usually assume that the cross-sectional units (e.g households, firms) are randomly sampled from a large population $\rightarrow$ i.i.d. across $i = 1, ..., n$.

At the same time, we will not make the i.i.d assumption within a unit i across time (i.e., for a given unit i), as it is typically implausible in most settings.

- example: income of household i in year t and year $t + 1$

Within- and between-Variation

Pooled OLS

We can write a linear panel data model as:

$$Y_{it} = \boldsymbol{X}_{it}\boldsymbol{\beta} + u_{it}, i = 1, ..., n; t = 1, ..., T.$$

$\boldsymbol{X}_{it}$ includes all regressors.
They might vary only across i (e.g. minimum drinking age), only across t (e.g. year dummies) or across both i and t (seat belt usage rate).

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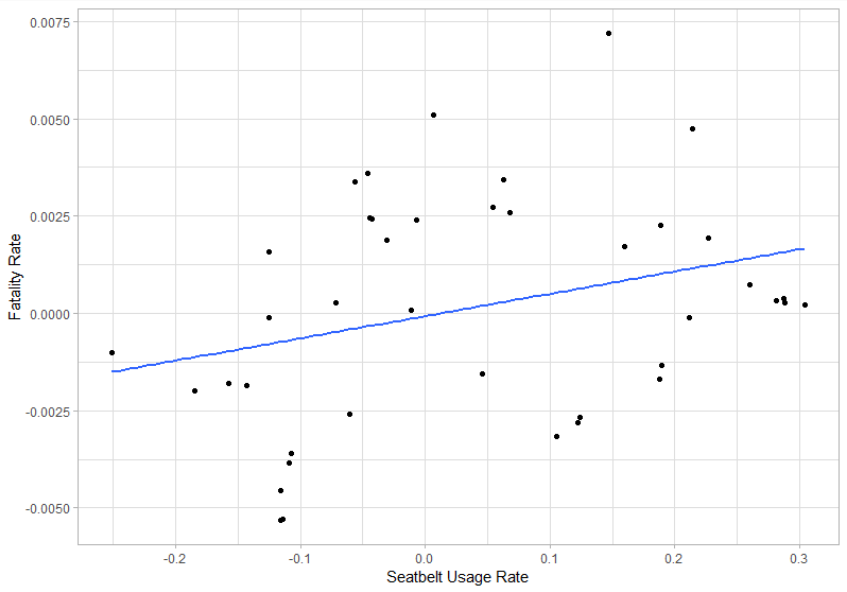
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Pooled OLS = estimate the parameters of the model by OLS

- treats the cross-section and time series dimensions of the data in the same way
- exploits *all* variation in the data, both within and between cross-sectional units.
- consistent under MLR.1 to MLR.4
  (MLR.2 implies: observations for different t are sampled independently).

Example: Pooled OLS



Within- vs. between-variation

Pooled OLS treats the whole data as one large (pooled) i.i.d. sample.

- often not justified as observations are not independent within units i across t.
- moreover, it does not exploit the specific advantages of panel data!

In a panel data set, we can decompose the total variation in our variables into

1. within variation: changes within each cross-sectional group i over time.
2. between variation: changes between cross-sectional groups

Example: seat belt usage rate varies *between* states, and *within* each state across years. minimum drinking age only varies between states. It is time-invariant.

The Panel Model

In the same way, unobservable determinants of Y_{it} in the error include

- time-invariant components: only differ between units i (e.g. geography, population's attitude towards seat belts)
- and components that vary between units i *and* within unit i over time (e.g. weather conditions)

Assuming the error term consists of two components, we can write:

$$Y_{it} = \boldsymbol{X}_{it}\boldsymbol{\beta} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

c_i unobserved time-invariant characteristics of unit i.

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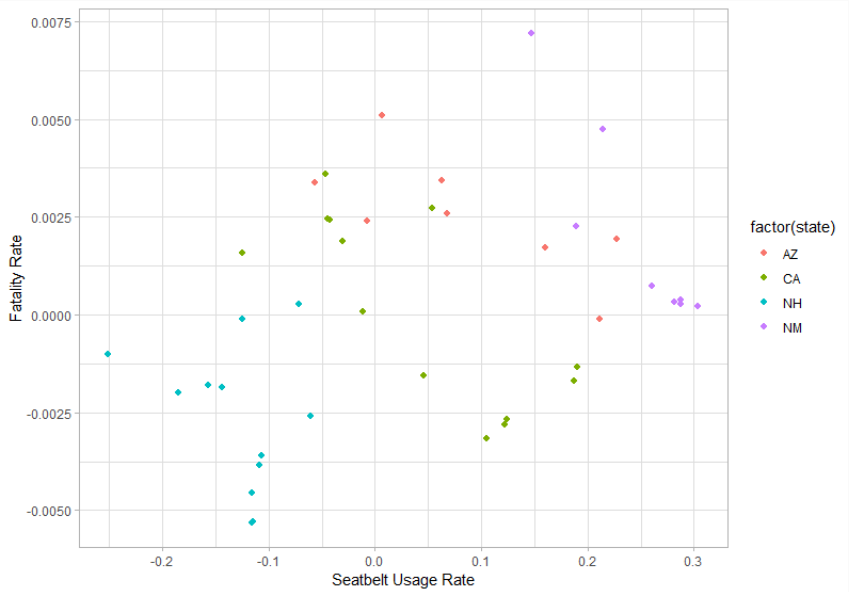
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with c_i **potentially correlated with included regressors:**

$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Within- and between-Variation



Panel Data and Time-Invariant Unobservables

c_i (= individual **fixed effect** or unobserved heterogeneity) is fixed over time.

With panel data, we can use estimators that exploit only the *within*−variation in the data, thereby eliminating all time-invariant factors!

This can help us address OVB by controlling for *all time-invariant* error components c_i.

Let's look at different ways to estimate the fixed effects model.

Panel Data Estimators

The Dummy Variable Regression

Option 1) explicitly add c_i to our regression by adding a dummy for each cross-sectional unit i.

Example: Using data at the state×year level, we can "control" for a state-dummy (all time-invariant unobservable characteristics at the state level, e.g. local laws).

$$Y_{it} = \boldsymbol{X}_{it}\boldsymbol{\beta} + \alpha_1 1\{state = i\} + ... + \alpha_n 1\{state = n\} + \epsilon_{it}, \;\; i = 1, ..., n; t = 1, ..., T.$$

Note: we either have to omit one dummy, or the constant.

`Keeping controls and the state fixed, a change in` X_1 `by 1 unit is associated with...`

$\rightarrow$ coefficients are estimated from variation for one specific i over time only ($within-$variation):

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The Dummy Variable Estimator

We can estimate the model with the cross-sectional-unit-dummies by OLS to obtain $\hat{\boldsymbol{\beta}}$.
$\rightarrow$ OLS is unbiased if $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$ (MLR.4 for this model).

Because all time-invariant unobservables are eliminated from ϵ_{it}, this is a much less strong assumption than $E[u_{it}|\boldsymbol{X_{it}}] = 0$.

Example: weather conditions much more likely to satisfy conditional mean independence than population's attitude towards seat belts.

When thinking about omitted variables that might be correlated with included regressors, we can ignore all time-invariant unobservables. Every conceivable time-invariant variable is accounted for in our analyses!

Pooled OLS vs. Dummy Variable Estimator

«include = F»= library(sandwich) library(lmtest) dat <- read.csv("seatbelts.csv") head(dat) @ «»= pooled <-
lm(fatality$_{r}$ate seatbelt$_{u}$sage, data = dat)coeftest(pooled, vcov. = vcovHC(pooled, "HC1"))

@

Pooled OLS vs. Dummy Variable Estimator

«»= include state dummies dvr <-
lm(fatality$_r$ate $seatbelt_u$sage + $factor(state), data = dat)coeftest(dvr, vcov. = vcovHC(dvr, "HC1"))$

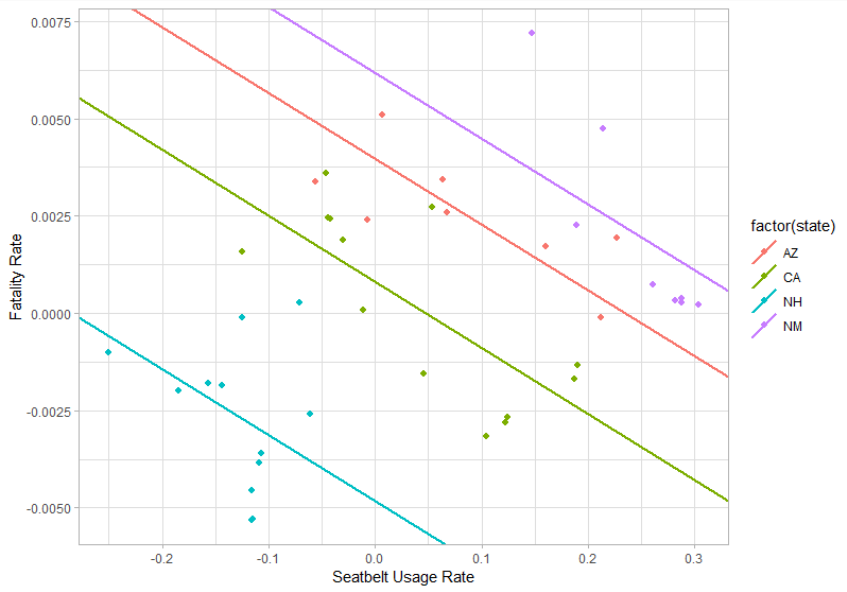
@

Interpretation

Pooled OLS:
Comparing seat belt usage rate and fatality rates *across states and years*, we find that an increase in the seatbelt usage rate by 0.1 (10 percentage points) is associated with an *increase* in the number of fatalities per million miles driven by .00057. (=2.9% of the sample mean).

Dummy variable estimation:
Comparing seat belt usage rate and fatality rates *within states over time*, we find that an increase in the seat belt usage rate by .1 is associated with a *decrease* in the number of fatalities per million miles driven by .0017. (=8.6% of the sample mean).

Illustration: Fixed Effects Estimation



Another Example

Climate change literature: how do rising temperatures affect human health and productivity?

Say we are interested in the effect of outdoor temperature (X_{it}) on exam performance (Y_{it}) among students in Italy.

We have a **panel** of students $i = 1, ..., n$ all across the country for multiple exams taken at dates $t = 1, ..., T$.

Just regressing Y_{it} on X_{it} (pooled OLS) might yield biased estimates, e.g. if students in the warmer South study harder because of bad economic prospects in the South.

$\rightarrow$ X_{it} correlated with u_{it} (e.g. effort)

Another Example

$$Y_{it} = \boldsymbol{X}_{it}\boldsymbol{\beta} + \underbrace{\alpha_1 1\{student = 1\}}_{c_1} + ... + \underbrace{\alpha_n 1\{student = n\}}_{c_n} + \epsilon_{it}, i = 1, ..., n; t = 1, ..., T.$$

Dummy variable regression allows to control for all time-invariant unobservables c_i, e.g. overall student effort and ability!

Effect of temperature will be estimated from variation for one specific student over exam dates (within-student variation).

First-Differences Estimator

Option 2) If $T = 2$, an alternative to the dummy variable approach is first-differencing:

Consider the models for the two time periods:

$$Y_{i1} = \boldsymbol{X}'_{i1}\boldsymbol{\beta} + c_i + \epsilon_{i1} \text{ and}$$
$$Y_{i2} = \boldsymbol{X}'_{i2}\boldsymbol{\beta} + c_i + \epsilon_{i2}$$

Differencing yields:

$$Y_{i2} \breve{} Y_{i1} = (\boldsymbol{X}'_{i2} - \boldsymbol{X}'_{i1})\boldsymbol{\beta} + (c_i \breve{} c_i) + (\epsilon_{i2} \breve{} \epsilon_{i1})$$
$$\text{or: } \Delta Y_i = \Delta \boldsymbol{X}_i \boldsymbol{\beta} + \Delta \epsilon_i$$

$\rightarrow$ The unobserved heterogeneity c_i drops out! (and also β_0)

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This is our "estimating equation".
We can estimate $\boldsymbol{\beta}$ by OLS by regressing ΔY_i on $\Delta \boldsymbol{X}_i$.
The estimator is unbiased if $E[\Delta \epsilon_i | \Delta \boldsymbol{X}_i] = 0$ (MLR.4 for this model).

Within Transformation

Option 3) The most common estimation approach is the *within-transformation*.

1. For all variables in your model, compute the time-mean for each unit i:
   $\overline{Y_i} = \frac{1}{T}\sum_{t=1}^{T} Y_{it}$, $\overline{X}_{1i} = \frac{1}{T}\sum_{t=1}^{T} X_{1it}$, ...

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2. For each observation, deduct the time means from the actual values. Obtain the *demeaned* variables $(Y_{it} - \overline{Y_i}), (X_{1it} - \overline{X}_{1i}), ..., (X_{kit} - \overline{X}_{ki})$

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3. Regress $(Y_{it} - \overline{Y_i})$ on $(X_{1it} - \overline{X}_{1i})$,..., $(X_{kit} - \overline{X}_{ki})$

Demeaned Equation:

$$\begin{aligned} Y_{it} - \overline{Y}_i &= (\beta_0 - \beta_0) + \beta_1(X_{1it} - \overline{X}_{1i}) + ... + \beta_k(X_{kit} - \overline{X_{ki}}) + (c_i - c_i) + (\epsilon_{it} - \bar{\epsilon}_i) \\ &= \beta_1(X_{1it} - \overline{X}_{1i}) + ... + \beta_k(X_{kit} - \overline{X}_{ki}) + (\epsilon_{it} - \bar{\epsilon}_i) \end{aligned}$$

c_i drops out because there is no *within-variation* in c_i!

Again, we can estimate this "demeaned equation" by OLS.

Within Estimator – Manual

«include = FALSE»= meansx <- as.data.frame(tapply(dat$seatbelt_u$sage, datstate, mean)) names(meansx) <- "means$_x$" $meansx$state <- rownames(meansx) meansx dat = merge(dat, meansx) dat$fatal_d emeaned$ = datfatality$_r ate$ − datmeans$_x demeanvariables$@

« »= meansy <- as.data.frame(tapply(dat$fatality_r ate$, datstate, mean)) compute time means names(meansy) <- "means$_y$" $meansy$state <- rownames(meansy) meansy time mean of dependent variable for each state dat = merge(dat, meansy, by = "state") dat$fatal_d emeaned$ = datfatality$_r ate$ − datmeans$_y demeandependendvariablesameforregressor(computationommited)dat$seatbelt$_d emeaned$ = datseatbelt$_u sage$ − datmeans$_x$@

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$$\text{«»= mod}_{w}ithin < -lm(fatal_{d}emeaned\ seatbelt_{d}emeaned, data = dat)coeftest(mod_{w}ithin, , vcov. = vcovHC(mod_{w}ithin, "HC1"))@We obtain the same slope coefficient estimate with the dummy variable and the within-transformation approaches.$$

Within Estimator with `fixest`

We can directly implement the within-estimator in R using the command `feols` from the package `fixest`.

```
install.packages("fixest")
library(fixest)
fe_mod <- feols(fatality_rate ~ seatbelt_usage | state, data = dat)
summary(fe_mod)
```

```
  Fixed-effects:         state:  4
  Standard-errors:   Clustered (state)

                     Estimate Std.      Error     t value     Pr(>|t|)
  seatbelt_usage       -0.016942       0.003851  -4.39955   0.021769 *
```

Syntax is similar to `estimatr::iv_robust`: here the **fixed effects come after the** |

The command directly provides **cluster-robust standard errors**: they account for dependence between observations across time within a cross-sectional group (state).

Limitations of Panel Data Estimators

Unlike pooled OLS, the within-estimator exploits only *within*-variation for each unit i, i.e. deviations from the unit-specific mean. (like the other approaches we considered)

We cannot include time-invariant regressors when using these estimators. They would drop out, because they are absorbed by the individual fixed effect.

Example: in a wage regression, we cannot estimate the slope coefficient on *gender*.

Even for variables with only little time-variation, estimating effects might not work well (e.g. education only varies among very young individuals.)

Practice

We are interested in the effect of union membership on the wage.

1. Open the dataset `unions.csv`, a multi-year panel of workers with person identifier *nr*.
2. Pooled OLS: Regress *lwage* (log wage) on *educ*, *black*, *hisp*, *exper* and its square, *union* and all year dummies: *d*82, *d*83, ...
3. What is the effect of union membership according to this model? Why might the coefficient be inconsistent?
4. What is the wage gap between white and black workers? How did wages change on average between 1982 and 1985?
5. Use `feols` to take individual fixed effects into account. How does the estimated effect of union membership change?
6. Why can you not estimate the black-white wage gap with the fixed effect estimator?
7. Keep only the years 1983 and 1987. Repeat the feols estimation. Then, add a regressor to estimate how the black-white gender gap *changed* between these two years.